ME46060 Engineering Optimization

Exercise 5

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Exercise 5

5.1 Finite difference gradients

At the design point $x = (0.024, 0.004)$, forward finite differencing is utilized to estimate the derivatives of the objective function and constraints with respect to the design variables. The accuracy of these calculated gradients is governed by two competing errors:

- **Truncation Error:** Dominant at large h due to the linear approximation of non-linear functions. As shown in Figure 1, this error appears on the right side of the plots where the gradient value begins to diverge from the actual derivative as the step size increases.
- **Round-off Error:** Dominant at small h due to machine precision limits. This is visible on the left side of the plots in Figure 1 as erratic oscillations or numerical noise occurring when the difference between the perturbed and original function values is too small for the computer to represent accurately.

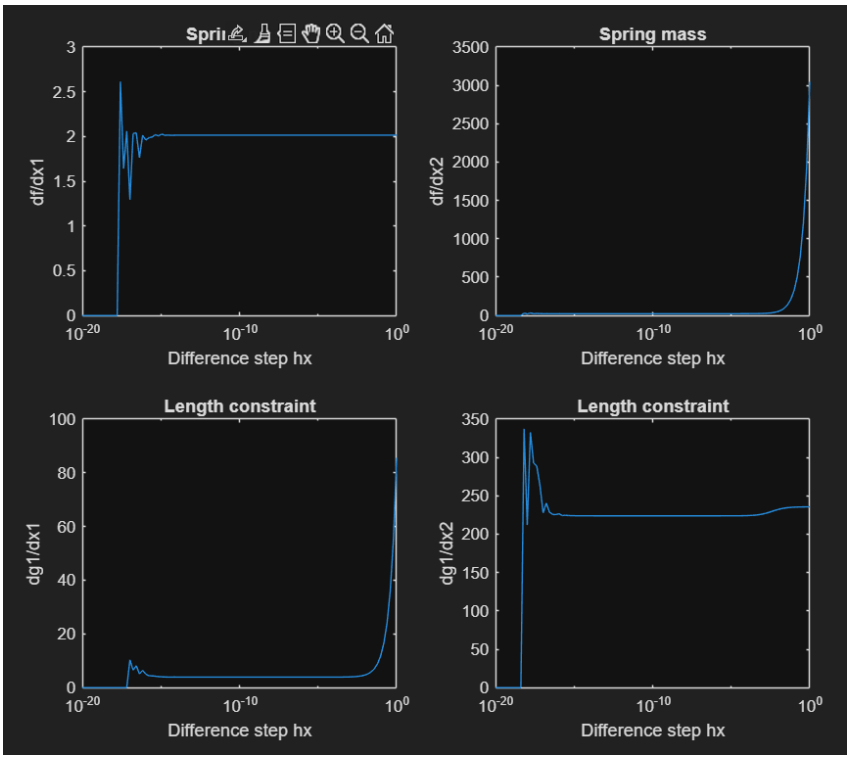

Figure 1: Finite difference gradients for spring mass and g_1 with default precision.

As seen in Figure 1, the usable range for the default sixteen-digit precision is approximately 10^{-12} to 10^{-4}.

By adding the expression $g(1) = str2double(num2str(g(1)))$, the constraint g_1 is limited to only five significant digits. As observed in Figure 2, this causes the round-off error to occur at much larger step sizes, specifically $h \approx 10^{-8}$. Consequently, the usable range for the length constraint g_1 is shifted and narrowed compared to the original precision.

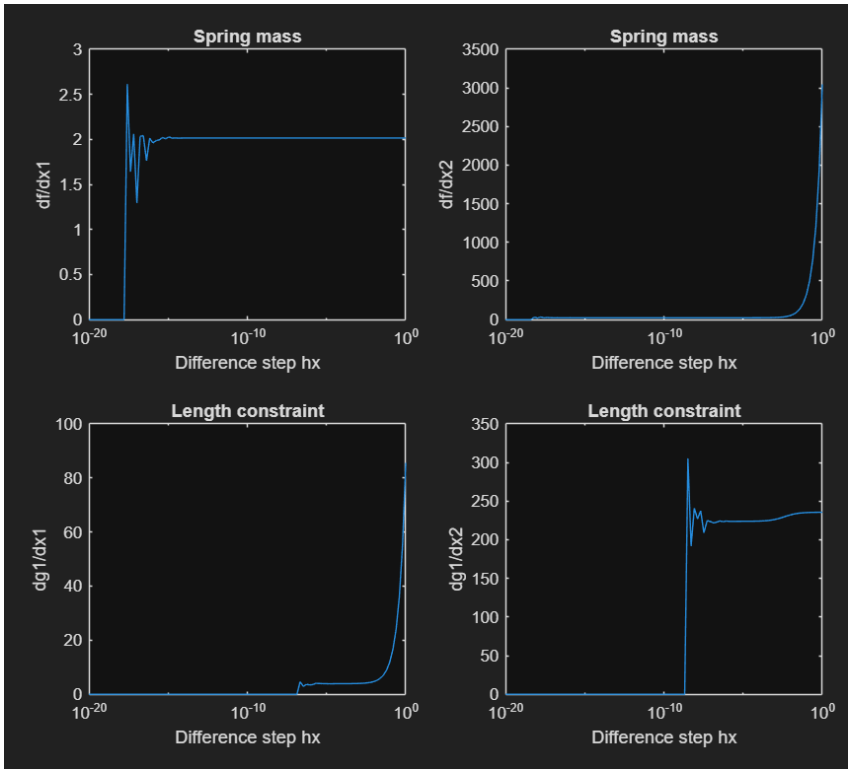

Figure 2: Gradients for g_1 with reduced precision (5 significant digits).

The gradients for the shear stress constraint (g_4) and frequency constraint (g_5) exhibit similar stability regions. However, as shown in Figure 3, the varying slopes of the truncation error plots at higher step sizes indicate that different physical constraints have different sensitivities to the design variables D and d.

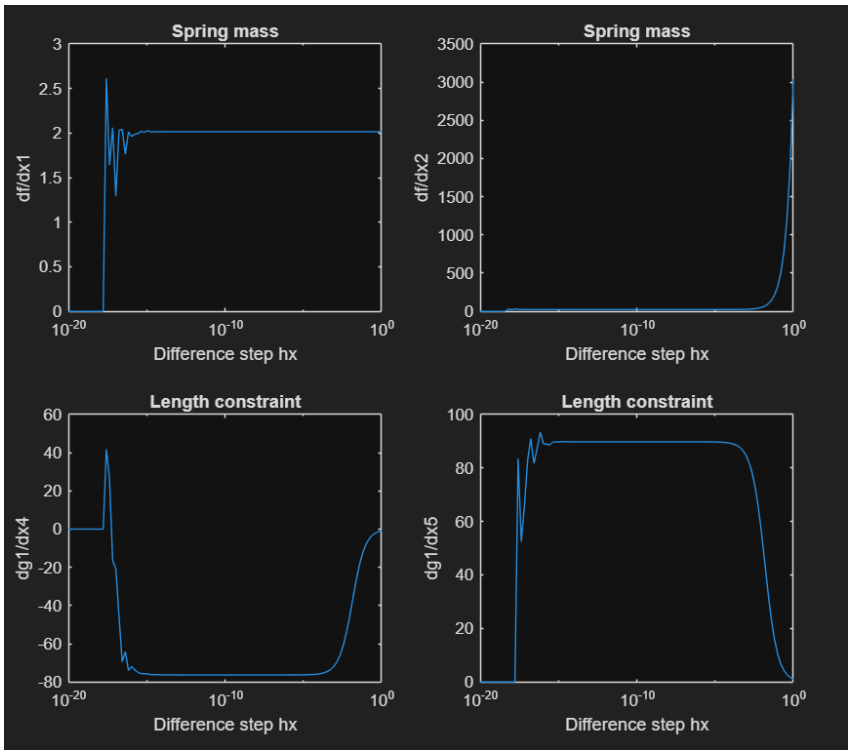

Figure 3: Finite difference gradients for constraints g_4 and g_5.

5.2 Karush-Kuhn-Tucker conditions

The Karush-Kuhn-Tucker conditions represent the necessary conditions for a constrained optimum. Point $x^* = [0.020522,\ 0.003520]$ is the constraint optimum of the valve spring optimization problem as given above. At this point, constraints $g_2,\ g_3$ and g_4 are active. A script *springw5ex2* is made to check the KKT conditions at x^*. This script works as follows:

- The values of the constraints $g_1(x)$ to $g_5(x)$ are computed. If $g_i(x) \approx 0$, the constraint is active.
- The gradients ∇f and ∇g_i are computed, where g_i are all active constraints.
- The system of linear equations $\begin{bmatrix}\nabla g_2 & \nabla g_3 & \nabla g_4\end{bmatrix} \cdot \begin{bmatrix}\mu_2 & \mu_3 & \mu_4\end{bmatrix}^T = -\nabla f$ is solved to find the Lagrange multipliers μ_i.
- The Lagrange multipliers are then checked if they are larger than 0. If this is the case, they meet the KKT condition for inequality constraints.

The results are shown in table 1.

Table 1: KKT check at x^*

Parameter	**Value**
$g_1(x^*)$	-0.131451
$g_2(x^*)$	-0.000016
$g_3(x^*)$	-0.000016
$g_4(x^*)$	-0.000037
$g_5(x^*)$	-0.296564
μ_2	0.017389
μ_3	0.017389
μ_4	0.061263

Table 2: KKT check at x_A and x_B

Parameter	x_A	x_B
$g_1(x)$	-0.000107	-0.000063
$g_2(x)$	-0.000012	-0.409150
$g_3(x)$	-0.000012	-0.409150
$g_4(x)$	-0.211516	-0.000090
$g_5(x)$	-0.032662	-0.276790
μ_1	-0.162909	-0.123176
μ_2	-0.005747	-
μ_3	-0.005747	-
μ_4	-	0.016390
μ_5	-	-
Stationary Residual	$4.69e-14$	0

These results indicate that indeed x^* is a valid constrained optimum because it satisfies all the required KKT conditions within numerical tolerances. The inactive constraints g_1 and g_2 are strictly negative and the active constraints are virtually 0. Additionally, the script found that μ_2 and μ_3 are identical. This is a direct result of g_2 and g_3 being dependent.

The script is then checked for the following 2 points:

- $x_A = [0.02462,\ 0.004035]$: The intersection of constraints g_2 (and g_3) and g_1.
- $x_B = [0.022251,\ 0.004075]$: The intersection of constraints g_4 and g_1

The results are shown in table 2. The numerical analysis of points x_A and x_B demonstrates that they do not satisfy the necessary KKT conditions for a constrained minimum. While both points achieve a low stationary resiudal, they fail the non-negativity test for Lagrange multipliers. At x_A all $\mu_i < 0$, but the larger magnitude of μ_1 indicates that this is the primary constraint making the point suboptimal. For x_B the lagrange multiplier $\mu_1 < 0$, hence this length constraint is the reason for sub-optimality. x^* is therefore the only verified constrained optimum among the points tested.